

Paper Title

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Abstract

[TODO: Write abstract here.]

Keywords

keyword1, keyword2, keyword3

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1 Introduction

[TODO: Motivate the problem: multi-language syntax conventions for LLM pretraining detection.]

[TODO: Summarize contributions:]

[TODO: Contribution 1]

[TODO: Contribution 2]

[TODO: Contribution 3]

2 Motivation

2.1 Problem Statement

[TODO: Describe the gap: existing MIA methods treat code as plain text and are language-agnostic.]

2.2 Observations

[TODO: Empirical observation: syntax tokens constitute a large fraction of tokens across languages (C, Java, JavaScript, etc.).]

3 Methodology

3.1 Preliminary

[TODO: Define membership inference attack (MIA) for code LLMs. Define syntax conventions.]

3.2 Multi-Language Syntax Conventions

[TODO: Describe how syntax conventions are collected for each target language (C, Java, JavaScript).] [TODO: Categorize conventions: data model, expressions, simple statements, compound statements.]

Data Model: [TODO: ...]

Expressions: [TODO: ...]

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Table 1: Benchmark Statistics

Language	Label	Files	Func.	Avg. LOC
C	Members	-	[TODO: N]	[TODO: X]
	Non-members	[TODO: N]	[TODO: N]	[TODO: X]
Java	Members	-	[TODO: N]	[TODO: X]
	Non-members	[TODO: N]	[TODO: N]	[TODO: X]
JavaScript	Members	-	[TODO: N]	[TODO: X]
	Non-members	[TODO: N]	[TODO: N]	[TODO: X]

Table 2: Syntax Convention Token Counts per Language

Language	Data Model	Expr.	Stmt.	Ratio
C	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X%]
Java	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X%]
JavaScript	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X%]

Simple Statements: [TODO: ...]

Compound Statements: [TODO: ...]

3.3 Syntax-Pruned MIA

[TODO: Phase 1: Token preprocessing.]

[TODO: Phase 2: Syntax-convention-based token pruning.]

$$X \in \{0, 1\} \quad (1)$$

[TODO: Phase 3: Member probability calculation.]

$$\text{SPP}(x) = \frac{1}{|X_1|} \sum_{i \in X_1} \log p(x_i) \quad (2)$$

4 Benchmark Construction

[TODO: Explain why existing benchmarks (WikiMIA, MIMIR) are insufficient for multi-language code MIA.]

Members. [TODO: Describe member collection: source dataset, languages, sampling strategy.]

Non-members. [TODO: Describe non-member collection: post-cutoff GitHub repos, verification process.]

Ratio Settings. [TODO: Evaluate under 1:1, 1:5, and 5:1 member-to-non-member ratios.]

5 Experiments

5.1 Setup

[TODO: Describe evaluated LLMs (e.g., Pythia, GPT-Neo, StableLM, GPT-J).] [TODO: Describe baselines: Loss, ZLib, Min-K, DC-PDD.] [TODO: Evaluation metric: AUROC.]

5.2 RQ1: Overall Performance

[TODO: Main results table across languages and models.]

Figure 1: Overview of ToolName

Table 3: AUROC (%) across languages and models (1:1 ratio)

Lang.	Method	Pythia 2.8B	GPT-Neo 2.7B	StableLM 3B	GPT-J 6B
C	Loss	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	ZLib	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	Min-K	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	DC-PDD	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	ToolName	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
Java	Loss	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	ZLib	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	Min-K	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	DC-PDD	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	ToolName	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
JS	Loss	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	ZLib	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	Min-K	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	DC-PDD	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
	ToolName	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]

5.3 RQ2: Robustness across Member/Non-member Ratios

[TODO: Results under 1:5 and 5:1 imbalanced settings.]

5.4 RQ3: Robustness across Code Length

[TODO: Performance breakdown by short / medium / long functions.]

5.5 RQ4: Ablation Study

[TODO: Effect of each syntax convention category on AUROC.]

6 Case Study

[TODO: Select representative LLMs and query with known copyleft-licensed functions (GPL, Apache, MIT).]

[TODO: Show evidence of memorized code across C / Java / JavaScript.]

[TODO: Discuss implications for copyright compliance.]

7 Related Work

Membership Inference Attacks. [TODO: Shadow model-based MIA [?]. Score-based MIA for LLMs: Loss, ZLib, Min-K [?], DC-PDD.]

MIA for Code. [TODO: GotCha [?], copyright-focused MIA [?], Wan et al. [?].]

LLM Training Data Transparency. [TODO: Pile [?], open-source license compliance, copyleft litigation.]

8 Conclusion

[TODO: Summarize: ToolName extends syntax-pruned MIA to multiple programming languages (C, Java, JavaScript), consistently outperforming baselines and providing evidence of copyleft code memorization in LLMs.]

A Ethics Statement

[TODO: Describe responsible use: research context only, no exploitation, adherence to responsible disclosure.]

B Acknowledgements

[TODO: Funding sources and reviewer thanks.]

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